MILITARY SPECIFICATION

DELAY ELEMENT, PA 500, FOR MINE M16

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the Delay Element For The M16A2 Antipersonnel Mine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation - Packaging, Methods of
MIL-P-223 - Powder, Black
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1345

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DRAWINGS

U.S. ARMY PICATINNY ARSENAL

8796352 - Delay Element, Mine: PA 500
9311219 - Box Set-up, Packing, Ammunition, for Delay Element, Mine: PA 500
9311220 - Packing and Marking for Box, Wirebound and Box, Metal, Ammunition M2A1 for Delay Element, Mine: PA 500

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Moisture Content of Black Powder. The moisture content of the black powder at the time of loading shall not exceed the percentage specified in Specification MIL-P-223.

3.3 Workmanship. All parts shall be fabricated and finished (or fabricated and loaded) in a thorough, workmanlike manner, and all manufacturing, processing and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.4 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 . Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or

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in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. - The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of 25 delay washers, 25 delay housings and 50 complete delay element assemblies. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

4.3.2 Inspections to be performed. - See MIL-A-48078, and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Delay Assembly and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Washer, Delay (Dwg. 8796372)				
	Examination for Defects	25		3.1	4.4.2.1
	Housing, Delay (Dwg. 8796371)				
	Examination for Defects	25		3.1	4.4.2.2
	Delay Element, Mine: PA500 (Dwg. 8796352)				
	Examination for Defects	25(a)		3.1	4.4.2.3
	Burning Time of Delay Assembly	50		3.1	4.5.2
	(a) The 25 used for Examination for Defects may be part of the sample of 50 to be tested for Burning Time.				

NOTES:

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Lot numbering, as required, shall be in accordance with MIL-STD-1168. In addition, inspection lots of Mine Assemblies shall contain:

Black powder of one interfix lot number from one manufacturer.

4.4.2 Examination. - (See MIL-A-48078). - Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Washer, Delay		8796372
			NEXT HIGHER ASSEMBLY 8796352

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height (max)		0.40%	3.1	Gage
102	Outside diameter (max)		0.40%	3.1	Gage
103	Hole diameter (min)		0.40%	3.1	Gage
104	Taper		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Housing, Delay				8796371
					NEXT HIGHER ASSEMBLY 8796352
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall height (max)		0.40%	3.1	Gage
102	Diameter of flash hole (min)		0.40%	3.1	Gage
103	Diameter of loading hole (min)		0.40%	3.1	Gage
104	Outside diameter of housing (max)		0.40%	3.1	Gage
105	Pitch diameter of threads (max)		0.40%	3.1	Gage
106	Wrench slots location and size		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Delay Element, Mine: PA500		8796352
			NEXT HIGHER ASSEMBLY
			8864746

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Delay charge missing		0.40%	3.1	Visual
102	Delay washer missing		0.40%	3.1	Visual
Minor					
201	Delay washer crimps less than 360 degrees		0.65%	3.1	Visual
202	Delay washer inverted		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Box, Set-up, After Loading and Prior to Assembling Top and Sealing Strip				9311219
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Any filler missing		100%	3.1	Visual
Major	None defined				
Minor					
201	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Box, Set-up, After Assembly of Top and Sealing Strip but Prior to Insertion into Sealed Bag				9311219
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Top damaged or improperly assembled		0.40%	3.1	Visual
102	Sealing strip missing, loose or improperly assembled		0.40%	3.1	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.1	Visual
202	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Box, Metal, Ammunition M2A1 Prior to Closing				9311220
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bags containing set-up boxes damaged		0.40%	3.1	Visual
102	Cushioning material incomplete or missing from around outer set-up boxes		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Box, Metal Ammunition M2A1 After Sealing				9311220
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged		0.40%	3.1	Visual
102	Marking missing, incomplete or illegible		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Box, Wood, Wirebound For Delay Element, Mine: PA500		9311220
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wires missing, broken, loose or omproperly fastened		0.40%	3.1	Visual
102	Box damaged		0.40%	3.1	Visual
103	Markings missing, incomplete or illegible		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48889 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Pallet Box (Alternate Pack)		9311220
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box improperly assembled		0.40%	3.1	Visual
102	Strapping missing, broken or loose		0.40%	3.1	Visual
103	Marking missing, incomplete or illegible		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.3	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Moisture Content of Black Powder. (See 3.2) Major Defect A. The moisture content of the black powder shall be determined at the time of loading not less than once for each eight hours of production, or portion thereof. Two grams of black powder shall be selected just prior to loading and the moisture content determined as specified in 4.5.1. Composite samples shall not be used in this determination. If the moisture content exceeds the requirement specified, the black powder shall be rejected. If this determination is made after loading has begun, components loaded therewith shall be rejected.

4.4.3.2 Burning Time of Delay Assembly - Detonator Assembly (see 8796352) Major Defect A - Eighty (80) delay assembly-detonator assemblies shall be selected from each lot for test. The test shall be performed as specified in 4.5.2. If two or more delay assemblies fail to comply with the applicable requirements, the lot shall be rejected. Any delay assembly whose delay time is lost due to equipment failure or error in procedure shall be replaced in order to determine if the lot complies with the delay time requirements.

4.4.3.3 Heat Seal Test of Sealed Bag (see dwg. 9311219) Major A Defect. - Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test Methods and Procedures.

4.5.1 Moisture Content of Black Powder. The moisture content of the black powder shall be determined as specified in Specification MIL-P-223.

4.5.2 Burning Time of Delay Assembly. - The delay assembly shall be tested for burning time using methods and equipment in accordance with 4.4.4.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - The delay elements shall be packaged in accordance with dwg. 9311219.

5.2 Packing and Marking. - The delay elements shall be packed and marked in accordance with dwg. 9311220.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.2 Submission of Inspection Equipment Designs for Approval. - (see MIL-A-48078. Submit equipment designs as required to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey, 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-0158

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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